

Package ‘tinydev’

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Title A Tiny R Package Development Toolkit

Version 0.1

Description A minimalistic R package development toolkit that provides essential functions for creating R packages with ease. 'tinydev' is to 'devtools' what 'tinytest' is to 'testthat', a lightweight alternative that focuses on simplicity and ease of use.

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tinydev-package	<i>A Tiny R Package Development Toolkit</i>
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Description

A minimalistic R package development toolkit that provides essential functions for creating R packages with ease. 'tinydev' is to 'devtools' what 'tinytest' is to 'testthat', a lightweight alternative that focuses on simplicity and ease of use.

check_cpp_compiler	<i>Check the C++ compiler</i>
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Description

Compile and load a minimal C++ shared library (a function equivalent to `function(a, b) a + b`) using R CMD SHLIB, to confirm that a working C++ compiler is available for building packages with compiled code (e.g. packages that use 'cpp4r'). Prints the compiler banner reported by R CMD SHLIB (e.g. using C++ compiler: 'g++ (GCC) 15.2.1 20250813') and the result of a test call into the compiled function.

Usage

```
check_cpp_compiler()
```

Value

Invisibly returns a list with `compiler` (the compiler banner reported by R CMD SHLIB) and `result` (the result of the test C++ call, which should be 5).

Examples

```
check_cpp_compiler()
```

clear	<i>Clear the console</i>
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Description

Clear the console screen. It works in interactive R sessions.

Usage

```
clear()
```

Value

Invisibly returns 'TRUE'.

license

*Add a license to a package***Description**

Copy a license file into the package's 'LICENSE.txt', update the 'License' field in 'DESCRIPTION', and add 'LICENSE.txt' to '.Rbuildignore'. 'license_bsd2()', 'license_bsd3()' and 'license_mit()' additionally write the templated 'LICENSE' stub file required by those licenses. Available functions: 'license_agpl' (AGPL-3.0), 'license_apache' (Apache 2.0), 'license_artistic' (Artistic 2.0), 'license_cc0' (CC0 1.0), 'license_gpl2' (GPL-2), 'license_gpl3' (GPL-3), 'license_lgpl2' (LGPL-2.1), 'license_lgpl3' (LGPL-3), 'license_bsd2' (BSD 2-Clause), 'license_bsd3' (BSD 3-Clause) and 'license_mit' (MIT).

Usage

```
license_agpl(pkgdir = NULL)
license_apache(pkgdir = NULL)
license_artistic(pkgdir = NULL)
license_cc0(pkgdir = NULL)
license_gpl2(pkgdir = NULL)
license_gpl3(pkgdir = NULL)
license_lgpl2(pkgdir = NULL)
license_lgpl3(pkgdir = NULL)
license_bsd2(pkgdir = NULL, year = NULL, copyright_holder = NULL)
license_bsd3(pkgdir = NULL, year = NULL, copyright_holder = NULL, organization = NULL)
license_mit(pkgdir = NULL, year = NULL, copyright_holder = NULL)
```

Arguments

`pkgdir` The path to the package directory. Defaults to `NULL`.

`year` Year (e.g., 2026). Only used by 'license_bsd2()', 'license_bsd3()' and 'license_mit()'.

`copyright_holder` Copyright holder (e.g., "package XYZ authors"). Only used by 'license_bsd2()', 'license_bsd3()' and 'license_mit()'.

`organization` Organization (e.g., "University of Surrey"). Only used by 'license_bsd3()'.

Value

Invisibly returns 'TRUE' if license was added successfully.

pkg_build

*Build R package***Description**

Build an R package into a tar.gz file. The built file will be created one level up from the package directory.

Usage

```
pkg_build(pkgdir = ".", document = TRUE, build_vignettes = TRUE, build_manual =
```

Arguments

```
pkgdir          The path to the package directory. Defaults to the current directory.
document        [logical] Generate documentation before checking.
build_vignettes [logical] Build vignettes. Defaults to TRUE.
build_manual    [logical] Build manual. Defaults to TRUE.
```

Value

Prints the path to the built tar.gz file if successful.

pkg_check	<i>Check R package</i>
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Description

Build and check an R package using R CMD check.

Usage

```
pkg_check(pkgdir = NULL, cran = TRUE, document = TRUE, manual = TRUE, vignettes
```

Arguments

```
pkgdir          The path to the package directory. Defaults to NULL.
cran            [logical] Run checks as if on CRAN (passes --as-cran).
document        [logical] Generate documentation before checking.
manual          [logical] Build the PDF manual (requires LaTeX).
vignettes       [logical] Build vignettes during check.
```

Value

Invisibly returns 'TRUE' if the package was built successfully.

pkg_check_docker *Check R package using Docker*

Description

Build and check an R package using R CMD check inside a Docker container. This uses R-Hub Docker images to provide a consistent checking environment. The package is built outside the container, then installed and checked inside. See <https://r-hub.github.io/containers/containers.html> for the details.

Usage

```
pkg_check_docker(pkgdir = ".", image = "ubuntu-gcc", document = TRUE, cran = TRUE)
```

Arguments

pkgdir	The path to the package directory. Defaults to the current directory.
image	[character] R-Hub Docker image name. One of: ubuntu-clang, ubuntu-gcc, ubuntu-next, ubuntu-release, atlas, clang-asan, clang-ubsan, clang21, clang22, donttest, gcc16, gcc-asan, lto, mkl, nold, nosuggests, rchk, valgrind.
document	[logical] Generate documentation before checking.
cran	[logical] Run checks with --as-cran flag.

Value

Invisibly returns 'TRUE' if the package check completed successfully.

pkg_clean *Clean R package binaries*

Description

Remove compiled binaries from the 'src/' directory of an R package. Only applies to packages with compiled C/C++ code and will remove files with extensions '.o', '.so', and '.dll'.

Usage

```
pkg_clean(pkgdir = ".")
```

Arguments

pkgdir	The path to the package directory. Defaults to the current directory.
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Value

Invisibly returns 'TRUE' if binaries were removed successfully.

pkg_copy_filtered *Copy package sources while respecting .Rbuildignore*

Description

Copy only the files of a package that would NOT be excluded by its '.Rbuildignore' file into a destination directory. Directories that match an ignore pattern are never recursed into (and so never listed), which avoids the cost of walking huge ignored trees (e.g. a 'dev/' folder with a JS 'node_modules' install) before handing the source off to 'R CMD build'/'R CMD check'.

Usage

```
pkg_copy_filtered(pkgdir, dest)
```

Arguments

pkgdir	The path to the package directory.
dest	The destination directory to copy the filtered sources into. Created if it does not already exist.

Value

Invisibly returns the destination path.

pkg_coverage *Check R package coverage*

Description

Check how much of an R package is tests by unit tests, vignettes and/or examples.

Usage

```
pkg_coverage(pkgdir = NULL, quiet = TRUE, type = "all")
```

Arguments

pkgdir	The path to the package directory. Defaults to NULL.
type	The type of coverage to calculate. One of "tests", "examples", "vignettes", or "all". Defaults to "all".
quiet	If TRUE, suppress output from the underlying covr::package_coverage() function. Defaults to TRUE.

Value

Invisibly returns 'TRUE' if the package was built successfully.

pkg_document	<i>Document R Package Functions</i>
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Description

Generate documentation for an R package using tinyroxygen.

Usage

```
pkg_document(pkgdir = NULL)
```

Arguments

pkgdir The path to the package directory. Defaults to NULL.

Value

Invisibly returns 'TRUE' if the documentation was generated successfully.

pkg_install	<i>Install R package</i>
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Description

Install an R package from a directory

Usage

```
pkg_install(pkgdir = ".")
```

Arguments

pkgdir The path to the package directory. Defaults to the current directory.

Value

Invisibly returns 'TRUE' if the package was installed successfully.

pkg_load	<i>Load package into the current R session</i>
----------	--

Description

Load all R code and compiled shared libraries from a package directory.

Usage

```
pkg_load(pkgdir = ".")
```

Arguments

pkgdir The path to the package directory. Defaults to the current directory.

Value

Invisibly returns a character vector of loaded R files.

pkg_register	<i>Register 'C++' Functions</i>
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Description

Register 'C++' functions in an R package that uses 'cpp4r'.

Usage

```
pkg_register(pkgdir = NULL)
```

Arguments

pkgdir The path to the package directory. Defaults to NULL.

Value

Invisibly returns 'TRUE' if the documentation was generated successfully.

pkg_template	<i>Start a new project with the cpp4r package template</i>
--------------	--

Description

Copy a minimal package template into a new directory.

Usage

```
pkg_template(path = NULL, pkgname = NULL)
```

Arguments

path	Path to the new project
pkgname	Name of the new package

Value

Invisibly returns 'TRUE' if the template was copied successfully.

Examples

```
# create a new directory
dir <- tempdir()
dir.create(dir)

# copy the package template into the directory
pkg_template(dir, "mynewpkg")
```

pkg_test	<i>Run tests</i>
----------	------------------

Description

Run tests for an R package using tinytest. Tests should be placed in the 'inst/tinytest' directory of the package. Any R files in that directory not starting with 'test' (e.g. 'helper.R') are sourced into the global environment before running the tests, making their definitions available to all test files.

Usage

```
pkg_test(pkgdir = ".")
```

Arguments

pkgdir	The path to the package directory. Defaults to the current directory.
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Value

Invisibly returns 'TRUE' if tests were run successfully.

<code>rbuildignore_add</code>	<i>Add a pattern to .Rbuildignore</i>
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Description

Add a pattern to the .Rbuildignore file in the current directory. If the file does not exist, it will be created. If the pattern already exists in the file, it will not be added again.

Usage

```
rbuildignore_add(pattern = NULL)
```

Arguments

<code>pattern</code>	Pattern to add (e.g., "LICENSE.txt" or "^LICENSE\\.txt\$").
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Value

Invisibly returns 'TRUE' if the pattern was added successfully or already exists in the file.

<code>tinydev-utils</code>	<i>Utility functions for tinydev</i>
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Description

Internal utility functions for tinydev package.

Usage

```
stop_if_not_package(path)
```

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